Sean Milliken

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Education

University of Pittsburgh Expected Graduation: April 2028

Bachelor of Science in Computer Science, Minor in Economics *Pittsburgh, PA*

- Coursework: Operating Systems, Data Structures and Algorithms I/II, Security and Privacy, Foundations of Machine Learning, Cloud Computing, Software Quality Assurance, Computer Organization and Assembly
- **GPA:** 3.5 / 4.0

Experience

Pitt AWS Cloud Innovation Center Jan 2026 – Present

Back-End Developer *Pittsburgh, PA*

- Designed and deployed a fully serverless AWS application to automate PA Program Accreditation.
- Built AWS Step Functions workflows that orchestrated concurrent LLM pipelines for document ingestion and accreditation standard mapping across 200+ compliance standards.
- Improved system reliability by implementing retry logic, exponential backoff, and dead-letter queues, enabling 44 Lambda functions to recover gracefully from LLM failures and API throttling.
- Developed AI extraction pipelines with pydantic-ai and AWS Bedrock to produce schema-validated outputs.
- Reduced accreditation compliance review time by over 60% by automating course-to-standard evidence.

Dunwoody Village June 2025 – Aug 2025

Information Technology Intern *Newtown Square, PA*

- Implemented real-time tracking and updates for over $100,000 worth of medical inventory.
- Supported residents and staff by troubleshooting Wi-Fi, Ethernet, VPN, and network access issues across a large residential campus.
- Assisted over 400 residents and staff with technology issues and resolving general IT tickets.
- Compiled over 3 years of invoice data in order to understand spending and overhead costs on unit renovations.

Projects

Blue-Chat | *Python*

- Developed a peer-to-peer bluetooth messaging application using RFCOMM sockets.
- Implemented client and server programs to establish and maintain connections between devices.
- Built bidirectional message transmission using UTF-8 encoding and persistent socket connections.

Crypto Price Predictor | *Python, HTML, CSS, JS*

- Developed for SteelHacks 2025 to predict the price of Crypto in the next 24 hours.
- Implemented using a Random Forest Classification ML algorithm trained on historical data from Yahoo Finance.
- Returns a prediction and confidence associated with the prediction.
- Uses a FastAPI to retrieve information from machine learning model.

Lyric Webscraper | *Python, HTTPX, Selectolax*

- Python web scraper to get song lyrics from Genius.com.
- Developed using HTTPX to process HTTP requests and uses Selectolax to parse the data.
- Extracted lyrics from structured HTML code.
- Handled inconsistent tag structures.

Technical Skills

Languages: Python, Java, C, C++, Typescript, HTML/CSS
Developer Tools: Git, AWS, Docker, Django, Linux
Libraries: Pandas, NumPy, NetworkX, Pydantic-AI